

A Character-Centric, Spoiler-Aware Storyline Visualization for Novel Reading

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Abstract

Storyline visualizations can help readers understand character interactions and narrative structure, but many existing systems are better suited to retrospective analysis than active reading. This project presents a character-centric, spoiler-aware storyline visualization for novels, using *The Hunger Games* trilogy as a case study. The system combines dual temporal views for chapter order and in-story chronology, spoiler-aware navigations, character-follow mode, semantic encodings for factions and relationships, and a persistent detail panel for narrative context. These features aim to support guided exploration for readers, teachers, and book clubs while avoiding premature plot disclosure. This paper describes the project context, design rationale, implementation, and a proposed evaluation focused on readability, character tracing, and spoiler safety. The prototype demonstrates how storyline visualization can be adapted from an analytical summary tool into a communication-oriented reading aid.

1 Introduction

Novels with large casts and shifting relationships can be difficult to follow, especially when readers want both structural overview and spoiler-safe exploration. While storyline visualizations offer a promising way to represent character co-presence over time, many existing systems are better suited to retrospective analysis than active reading. This project explores how storyline visualization can be adapted into a communication-oriented reading aid for fiction.

Contributions. This project makes the following contributions:

- a spoiler-aware interaction model for fiction exploration;
- a character-follow mode for egocentric narrative tracing;
- dual chapter-order and in-story chronology views;
- semantic encodings and a persistent detail panel for guided interpretation.

2 Background and Related Work

Storyline visualizations represent temporal interactions as lines that converge and diverge over time. Tanahashi and Ma formalized this design space and emphasized readability concerns such as reducing unnecessary wiggles, crossings, and wasted space while keeping interacting entities visually adjacent [1]. My project adopts this storyline metaphor as its visual backbone, using character trajectories and grouped interaction sessions while shifting the emphasis from layout optimization alone toward reader-oriented exploration.

Narrative visualization research further shows the

value of comparing multiple temporal structures, pairing overview with contextual detail, and supporting guided interpretation rather than free exploration alone [2, 3]. Story Curves distinguishes narrative order from story order and augments the main view with metadata such as characters, place, and time [4]. This directly motivates the chapter order versus in-story order toggle in my prototype, as well as the use of linked contextual detail. Story Ribbons similarly points toward richer literary interfaces with detail overlays and explanation-on-demand, although my system does not implement automated extraction [5].

Character-centric interaction is motivated by EgoLines, which supports egocentric analysis of dynamic networks by highlighting one focal actor while preserving surrounding context through filtering and coordinated interaction [6]. This idea informs my character-follow mode, where users can trace a selected character through the story.

Finally, DataToon suggests that communication-oriented visualizations benefit from annotation, progressive reveal, and explicit support for interpretation rather than only dense free exploration [2, 7, 8]. My project adopts that lesson through spoiler-aware progression, a persistent detail panel, and planned guided callouts. Together, these systems position the prototype at the intersection of storyline visualization, narrative exploration, egocentric interaction, and communicative design [3, 9].

3 Design

3.1 Data and Task Abstraction

The primary items in the system are *scenes* (or interaction sessions) and *characters*. Each scene contains participat-

ing characters and optional metadata such as chapter index, story-time position, location, faction or group labels, relationship tags, and a short narrative summary. The resulting dataset combines a temporal table of scene-level records with a relational structure over characters. Following the general storyline metaphor, scenes connect character trajectories over time while preserving both individual traceability and interaction readability [1].

A key modeling decision is to distinguish between *chapter order* and *in-story chronology*. Story Curves motivates separating narrative order from story order, since the order in which events are told may differ from the order in which they occur [4]. My prototype therefore stores both chapter position and story-time position, allowing users to compare reader-facing sequence with world-time sequence. The main tasks are tracing a focal character, inspecting co-presence within a scene, comparing chapter order with in-story chronology, identifying relationship or alliance shifts, and retrieving narrative detail for a selected event. These tasks combine global overview with local interpretation and motivate the use of both character-centric interaction and persistent contextual detail [5, 6, 8].

3.2 Audience and Usage Scenario

The target audience includes readers, teachers, and book clubs. Readers may want help following a character or recalling where a relationship changed, while teachers and discussion groups may want a shared visual aid for narrative structure and character dynamics. A central challenge is that not all users have finished the story, so retrospective summaries can easily introduce spoilers. The system therefore adopts a guided exploration scenario in which users begin in chapter order, set their current reading progress, and explore only the visible portion of the story. Once the story is marked as completed, they can unlock in-story chronology and inspect the narrative from a broader structural perspective. This usage model emphasizes spoiler-safe exploration, coordinated detail, and character-follow interaction rather than large-scale literary analysis.

3.3 Visual Encoding Design

The visualization adopts the storyline metaphor as its primary visual idiom. Characters are represented as line trajectories moving from left to right over time, while shared scenes are represented as grouped interaction regions that bundle co-present characters. This design follows the core structure of storyline visualization and prioritizes traceability, adjacency of interacting lines, and reduction of unnecessary crossings and deviations [1]. Horizontal position encodes temporal progression. In chapter mode, the x-axis follows reading order, while in story mode it follows in-story chronology [4]. Vertical position encodes character lanes, giving each character a stable identity anchor that supports reader-oriented tracking.

Additional semantic attributes are layered using visual channels. Color is used primarily for faction or group identity, shared interaction regions indicate co-presence, and opacity or styling supports focus and semantic distinction. In particular, the de-emphasis of unrelated lines in character-follow mode upon hovering allows the chart to shift from a global overview to a more local reading, following the character-centric logic of EgoLines [6]. Text labels and the detail panel provide contextual information that would be difficult to encode directly in the chart, drawing on the broader idea that narrative visualization benefits from pairing overview with richer metadata and explanation [4, 5].

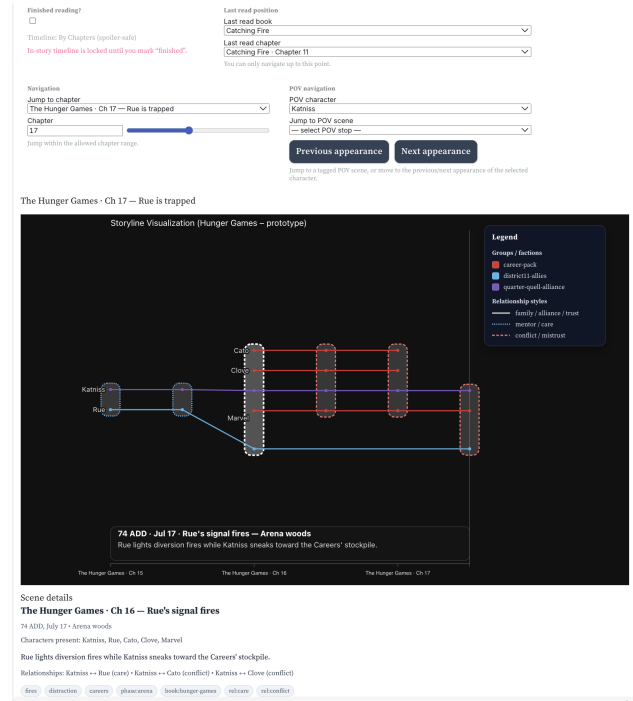


Figure 1: Prototype interface showing character trajectories, grouped interaction sessions, spoiler-aware controls, legend, and persistent detail panel.

3.4 Interaction Design

The system includes four main interaction mechanisms: spoiler-aware mode, character-follow mode, timeline switching, and a persistent detail panel. Together, these interactions shift the system from passive viewing toward guided narrative exploration.

Spoiler-aware mode adapts the visualization to the reader's current progress by restricting what can be explored beyond that point, while the in-story chronology remains locked until the story is marked as completed. Character-follow mode supports egocentric exploration by letting users select a focal character and move between appearances, following the actor-centered logic of EgoLines [6]. The timeline toggle supports comparison between chapter order and in-story chronology, adapting the distinction between narrative order and story order

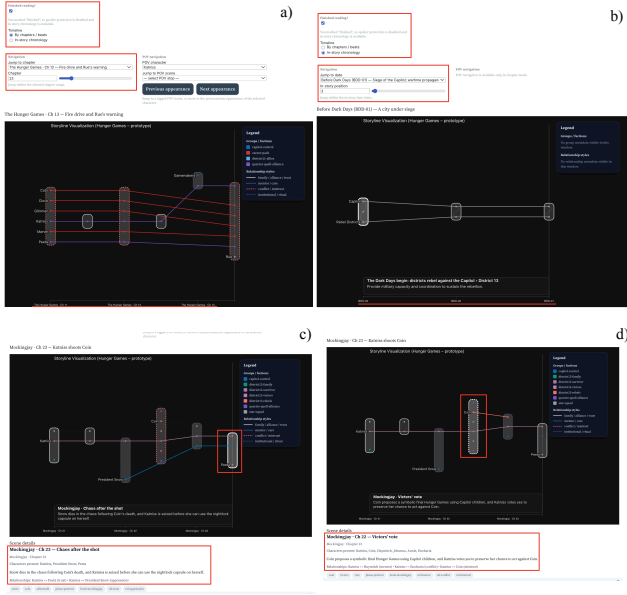


Figure 2: Timeline toggle and persistent contextual detail. **(a) and (b)**: Switching between chapter order and in-story chronology reorders the same narrative structure to support comparison between reading sequence and world-time sequence. **(c) and (d)**: Clicking a scene updates the persistent detail panel, allowing users to inspect summary text and related narrative metadata without relying only on transient hover text. Red annotations highlight the relevant control and view regions.

emphasized in Story Curves and drawing on broader ideas of temporal navigation in dynamic visualizations [4, 10]. Finally, the persistent detail panel provides stable contextual information for the selected chapter, scene, or character, reducing reliance on transient hover text and supporting interpretation through linked narrative metadata [4, 5, 8]. Together, these interactions support reading, teaching, and discussion workflows rather than retrospective summary alone. Figures 2 and 3 illustrate these interaction mechanisms through representative before/after states, highlighting the controls and view regions that change during timeline switching, detail inspection, character-follow navigation, and spoiler-aware progression.

4 Implementation

The prototype was implemented in Observable using D3. It reads structured scene-level narrative data derived from *The Hunger Games* and renders character trajectories, grouped interaction sessions, a legend, and linked detail views in a coordinated SVG-based interface.

The input dataset is organized around scenes or interaction sessions. Each record includes chapter position and, when available, story-time position, along with participating characters and optional metadata such as

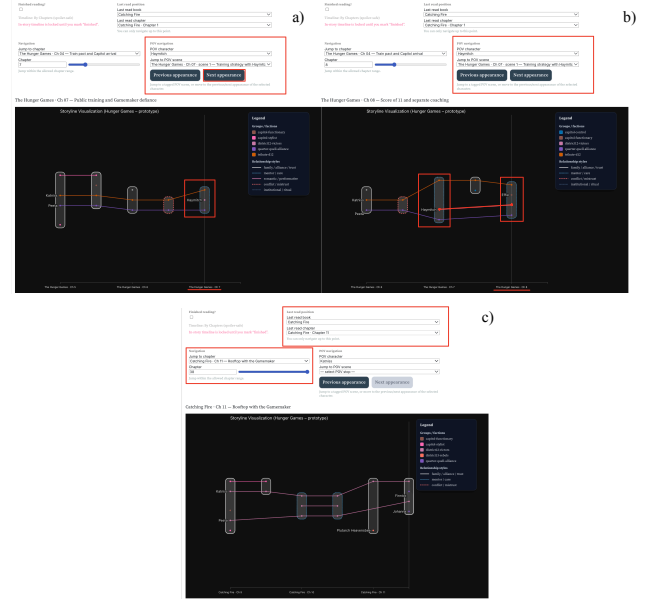


Figure 3: Character-follow / POV navigation and spoiler-aware progression. **(a) and (b)**: Selecting a focal character highlights that trajectory and supports previous/next appearance navigation while preserving surrounding narrative context. **(c)** Spoiler-aware mode restricts exploration based on the reader's current progress and keeps in-story chronology locked until completion. Red annotations highlight the relevant control and chart regions.

faction labels, relationship tags, location, and summary text. This representation supports both chapter-order and in-story chronology views within a single visualization pipeline. Several semantic attributes, such as faction and relationship tags, were manually curated.

Interaction state is managed through shared Observable variables, including timeline mode, reading progress, current selection, and spoiler restrictions. These states coordinate the storyline view, control bar, and detail panel. The implementation supports four main features: spoiler-aware navigation, character-follow mode, switching between chapter order and story chronology, and a persistent detail panel for contextual inspection.

The current system prioritizes design exploration over automation. Scene segmentation and semantic tagging are still manually prepared, so the prototype should be understood as a medium-scale research system focused on interaction design and reading workflows rather than large-scale automatic literary analysis.

5 Proposed Evaluation

A lightweight evaluation could compare the prototype against static chapter summaries and a non-interactive storyline view. The goal would be to test whether the added interactions such as spoiler-aware progression, character-follow mode, timeline switching, and the per-

sistent detail panel can improve navigation and interpretation.

Participants could perform tasks such as locating a character’s next appearance, identifying co-present characters in a selected scene, comparing chapter order with in-story chronology, or finding where a relationship shift occurs. These tasks reflect the system’s main intended uses, which are character tracing, scene inspection, temporal comparison, and contextual interpretation.

Measures could include task completion time, correctness, number of interactions, perceived clarity, and confidence about avoiding spoilers. A small within-subject study with readers or classmates would be practical for a course-scale evaluation and would also allow brief qualitative feedback on which interactions were most useful.

6 Discussion and Future Work

This project suggests that storyline visualization for fiction should support partial knowledge rather than assume full completion. Many existing narrative visualizations are most useful after a reader already knows the story, but active reading, teaching, and book-club discussion often happen under incomplete knowledge. The spoiler-aware interaction model in this prototype therefore points to a broader design implication, in which fiction-oriented visualizations should account for reader progress as part of the interface, not merely as an external condition.

A second lesson is that coordinated detail is essential for narrative interpretation. A storyline chart can provide structural overview, but it is often insufficient on its own for understanding why an interaction matters, what changes in a scene, or how a relationship evolves. This is consistent with systems such as Story Curves and Story Ribbons, which pair overview representations with richer contextual information, and with DataToon, which emphasizes communicative support through annotation and guided explanation. In this project, the persistent detail panel plays that role at a modest scale by reducing dependence on transient hover text and making scene-level interpretation more stable. [4, 5, 8]

The current prototype also has several limitations. First, the dataset depends on manual scene segmentation and manual semantic tagging, including faction and relationship labels. This limits scalability and introduces subjectivity into the representation. Second, dense scenes remain visually challenging: as more characters and relationships are shown in a limited space, clutter increases and the legibility goals identified in storyline visualization become harder to maintain. Third, the current implementation has only been tested as a medium-scale design prototype and has not yet been validated through a formal user study. These limitations mean that the system should be understood primarily as a design exploration rather than a fully general narrative-analysis tool. A small secondary prototype based on *Inception* also suggests that the scene-and-character abstraction can extend

to narratives with more simultaneous interaction bands and layered temporal structure, although that direction remains preliminary.

Future work could proceed in several directions. The most immediate step is lightweight user testing with readers or classmates to evaluate readability, character tracing, spoiler safety, and the usefulness of coordinated detail. A second direction is richer communicative support through annotations, guided callouts, or preset exploration paths, drawing more directly on the lessons of communication-oriented systems such as DataToon. A third direction is scaling the approach to more complex novels or serialized narratives, which would likely require improved layout strategies, stronger filtering mechanisms, and partial automation of scene extraction and tagging. More broadly, this project could be extended toward a reusable framework for reader-safe narrative visualization rather than a single case-study prototype.

7 Conclusion

This project shows how storyline visualization can be adapted from a retrospective analytical tool into a guided, communication-oriented reading aid for fiction. Using *The Hunger Games* trilogy as a case study, I designed and implemented a prototype that combines spoiler-aware navigation, character-follow interaction, dual temporal views, semantic encodings, and a persistent detail panel for contextual inspection. Together, these features aim to support readers, teachers, and book clubs who need both structural overview and scene-level interpretation without assuming complete knowledge of the story.

More broadly, the project suggests that narrative visualization for fiction benefits from treating reading progress, character perspective, and contextual explanation as first-class design concerns. Rather than presenting the entire story only as a completed object for retrospective analysis, the prototype supports a more guided and reader-centered form of exploration. Although the current system remains limited by manual data curation, subjective semantic tagging, and medium-scale scope, it offers a concrete direction for future work on reader-safe and communication-oriented narrative visualization.

References

- [1] Y. Tanahashi and K.-L. Ma, “Design considerations for optimizing storyline visualizations,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 12, pp. 2679–2688, 2012.
- [2] E. Segel and J. Heer, “Narrative visualization: Telling stories with data,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 16, no. 6, pp. 1139–1148, 2010.
- [3] R. Kosara and J. Mackinlay, “Storytelling: The next step for visualization,” *Computer*, vol. 46, no. 5, pp. 44–50, 2013.
- [4] N. W. Kim, B. Bach, H. Im, S. Schriber, M. Gross, and H. Pfister, “Visualizing nonlinear narratives with story curves,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 1, pp. 595–604, 2018.
- [5] C. Yeh, T. Menon, R. S. Arya, H. He, M. Weigel, F. Viégas, and M. Wattenberg, “Story ribbons: Reimagining storyline visualizations with large language models,” *arXiv preprint arXiv:2508.06772*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.06772>
- [6] J. Zhao, M. Glueck, F. Chevalier, Y. Wu, and A. Khan, “Egocentric analysis of dynamic networks with EgoLines,” in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, ser. CHI ’16. New York, NY, USA: ACM, 2016, pp. 5003–5014.
- [7] J. Hullman and N. Diakopoulos, “Visualization rhetoric: Framing effects in narrative visualization,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 17, no. 12, pp. 2231–2240, 2011.
- [8] N. W. Kim, N. H. Riche, B. Bach, G. Xu, M. Brehmer, K. Hinckley, M. Pahud, H. Xia, M. J. McGuffin, and H. Pfister, “Datatoon: Drawing data comics about dynamic networks with pen + touch interaction,” in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, ser. CHI ’19. New York, NY, USA: ACM, 2019, pp. 1–12.
- [9] F. Hohman, M. Conlen, J. Heer, and D. H. Chau, “Communicating with interactive articles,” *Distill*, vol. 5, no. 9, 2020. [Online]. Available: <https://distill.pub/2020/communicating-with-interactive-articles/>
- [10] B. Bach, E. Pietriga, and J.-D. Fekete, “Graphdiaries: Animated transitions and temporal navigation for dynamic networks,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 20, no. 5, pp. 740–754, 2014.